Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/26/2005
PI Name:	Meiburg, Eckart H. Ph.D.		
Project Title:	THE DYNAMICS OF MISCIBLE FLUIDS: A SPACE EXPERIMENT.		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
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Zip Code:	93106	**Congressional District:**	23
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	This project represents the computational component of a joint experimental and computational investigation into the dynamics of miscible interfaces. Within the computational component, it is proposed to carry out high resolution spectral element simulations that elucidate in significantly increased detail the role of 1) nontraditional Korteweg stresses and 2) non-vanishing divergence in miscible fluid flows. The geometry to be considered is the capillary tube also employed in the experiments. The proposed research aims at obtaining fundamental insight into the importance of Korteweg stresses and effects of non-vanishing divergence. For this purpose we will continue the preliminary numerical simulations for different fluid pairs, with corresponding density differences and estimated Korteweg stress coefficients. Over the next year, these simulations will be analyzed in significantly more detail than we have been able to obtain so far. This will be accomplished by employing high-accuracy spectral element methods with high resolution. By comparing the results of such simulations with those obtained for conventional systems (i.e., no Korteweg stresses and		

Task Description:	divergence free velocity field), we can obtain estimates for the importance of these effects in different miscible fluid combinations. In this way, the simulations will aid in the selection of optimally suited fluid pairs for the planned space experiments. Also, the discrepancies observed between the different simulations will guide experimental observations of flow features caused by the mechanisms under investigation. This research has a practical motivation, since the improved understanding of the dynamics of multiphase porous media flows is a pre-requisite for progress in the fields of enhanced oil recovery, fixed bed regeneration, hydrology, filtration etc. From stability theory we know that if in a porous environment the displacing fluid is less viscous than the displaced fluid, the unfavorable mobility profile leads to the well known fingering instability. The displacing fluid subsequently channels through the displaced zone, leaving behind a layer of the more viscous fluid at the walls, thereby reducing the efficiency of the displacement process. When the fluids are immiscible, the interfacial tension (or the Capillary number, which is a measure of the ratio of viscous to surface tension forces) determines the most unstable wavelength of the fingering instability as well as the dynamics of the fingers, as they evolve. When the liquids are miscible, a meniscus between the two is absent, and the interfacial tension tends to zero. Thus, at first sight, the dynamics appear to be simpler and are determined by the relative importance of convection and diffusion. However, the comparison of our ground based experiments and corresponding numerical simulations suggests that a number of secondary effects become important and influence the dynamics. The non-vanishing divergence of the velocity field and the Korteweg stresses, discussed earlier, are two such phenomena. In addition, the diffusion tensor at the interface may be anisotropic, and may depend on the local velocity field. While it is true that a large effective interfacial tension is unlikely to exist at a miscible interface, Joseph and coauthors have approached the stability problem based on the idea of such an effective interfacial tension. This approach was motivated by the observation that the form of miscible drops of one fluid moving in another is similar to that of immiscible fluids with a large interfacial tension. However, while it is easily possible to make such qualitative comparisons, it is much harder to establish them on sound quantitative footing. It is clear from the above discussion that miscible interfaces are rich in terms of dynamical phenomena that might influence its behavior. One output from this project would be an attempt to establish the proper form of the set of governing equations of motion for such flows. It is expected that these equations will be a modified form of the conventional Stokes equations, with additional or altered terms that allow for nonsolenoidal effects and Korteweg stresses. Under some circumstances, i.e. for some parameter regimes, we hope to be able to generate a suitable boundary condition to be used at a miscible interface, which will be compared with other approaches taken in the past.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Over the last year, we considered a vertically arranged Hele-Shaw cell in which a heavier fluid is placed above a lighter one. The two fluids are miscible with each other in all proportions. The effects of Korteweg stresses on this configuration are analyzed by means of a linear stability analysis of the three-dimensional Stokes equations. The goal is to perform comparisons with corresponding experimental data, in order to obtain quantitative estimates of the Korteweg stress coefficients. The growth rates of both the two- and the three-dimensional instability modes are seen to be reduced by these stresses. In that sense, the Korteweg stresses act like an `effective surface tension.' Other effects of these stresses, however, cannot be explained by a simple surface tension analogy, as they are linked to the finite thickness of the miscible interface region. For sufficiently large Korteweg stresses and fairly thin interfaces, the largest growth rates are observed at intermediate values of the Rayleigh number. This behavior, which is in contrast to conventional Stokes results, can be explained in terms of the damping effect of the Korteweg stresses at large Rayleigh numbers. Interestingly, experiments by \cite{Fernandezetal01} indicate a similar trend, which may indicate a certain influence of Korteweg stresses. The three-dimensional Stokes results are furthermore compared with their counterparts for the gap-averaged Hele-Shaw equations. It is found that the presence of Korteweg stresses leads to an improved agreement between the two sets of results. However, depending on the specific parameter values, the quantitative disagreement can still be substantial. We furthermore observe that three-dimensional Stokes results without Korteweg stresses can be reproduced, within limits, by introducing an empirical surface tension like parameter into the Hele-Shaw equations. However, for low Rayleigh numbers, the Hele-Shaw equations predict an unphysical increase in the growth rate with the Korteweg parameter.
Bibliography Type:	Description: (Last Updated: 04/27/2010)
Articles in Peer-reviewed Journals	Wilhelm, D.; Meiburg, E. "Three-Dimensional Spectral Element Simulations of Variable Density and Viscosity, Miscible Displacements in a Capillary Tube" Journal of Fluid Mechanics, submitted, 2004. , Jan-2004
Articles in Peer-reviewed Journals	Chen, C.-Y.; Meiburg, E. "Miscible Displacements in Capillary Tubes: Effect of a Preexisting Wall Film" Physics of Fluids , Feb-2004
Articles in Peer-reviewed Journals	Goyal, N.; Meiburg, E. "Miscible Displacements in Capillary Tubes: Effect of a Preexisting Wall Film" J. Fluid Mech, submitted , Jan-2004